

Teaching Exam

Subject – Computer Science

Topic – DBMS



Lecture No.- 8

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Topics to be Covered

Topic



- 1:- Normalization ✓
- 2:- 1- NF (Normal Form) ✓
- 3:- Closer method to find candidate key
- 4:- Some example related to closer method



Topic : Normalization

Normalization is the process of organizing the data and the attributes of a database. It is performed to reduce the data redundancy in a database and to ensure that data is stored logically. Data redundancy in DBMS means having the same data but at multiple places. It is necessary to remove data redundancy because it causes anomalies in a database which makes it very hard for a database administrator to maintain it.

Row Wise Redundancy:-

SID AGE
✓ 1 20
// 1 21

Composite Key

SID	SNAME	AGE
1	RAVI	20
2	SURYA	23
1	RAVI	21



Topic : Normalization

PK

SID	SNAME

FK PK

SID	CID	CNAME
2	2	DBMS
2	2	JAVA
3	2	C

PK

CID	FID	FNAME	SALARY
1	1	John	
2	2	Ravi	

COLLEGE MANAGEMENT

CM

STUDENT

COURSE

FACULTY

Column wise
Redundancy:-

Delete table
CM where
SName='Surya'

SID	SNAME	CID	CNAME	FID	FNAME	SALARY
1	RAM	C1	DBMS	F1	JOHN	30000
2	SURYA	C2	JAVA	F2	RAVI	40000
3	NITIN	C1	DBMS	F1	JOHN	30000
4	AMRIT	C1	DBMS	F1	JOHN	30000



Topic :Normalization

Why Do We Need Normalization?

As we have discussed above, normalization is used to reduce data redundancy. It provides a method to remove the following anomalies from the database and bring it to a more consistent state:

A database anomaly is a flaw in the database that occurs because of poor planning and redundancy. *→ problem*

- ✓ 1.-**Insertion anomalies**: This occurs when we are not able to insert data into a database because some attributes may be missing at the time of insertion.
- ✓ 2.-**Updation anomalies**: This occurs when the same data items are repeated with the same values and are not linked to each other.
- ✓ 3.-**Deletion anomalies**: This occurs when deleting one part of the data deletes the other necessary information from the database.



Topic : 1-NF

If a relation contains a composite or multi-valued attribute, it violates the first normal form, or the relation is in the first normal form if it **does not contain** any composite or multi-valued attribute. A relation is in first normal form if every attribute in that relation is single-valued attribute.

RAHUL Gupta

A relation (table) is said to be in First Normal Form (1NF) if:

- All the attributes (columns) contain only atomic (indivisible) values.
- Each column contains values of a single type.
- Each record (row) is unique, meaning it can be identified by a primary key.
- There are no repeating groups or arrays in any row.



RAHUL
Multivalued





Topic :1-NF

It is Not in 1 NF

ROLL NO	NAME	COURSE
1	RAVI	C/C++
2	HARI	JAVA
3	OHM	C/DBMS

METHOD- 1ST

COMPOSITE KEY (ROLL NO+COURSE), HERE THERE IS NO PRIMARY KEY

ROLLNO	COURSE	NAME
1	C	RAVI
1.	C++	RAVI
2	JAVA	HARI
3	C	OHM
3	DBMS	OHM

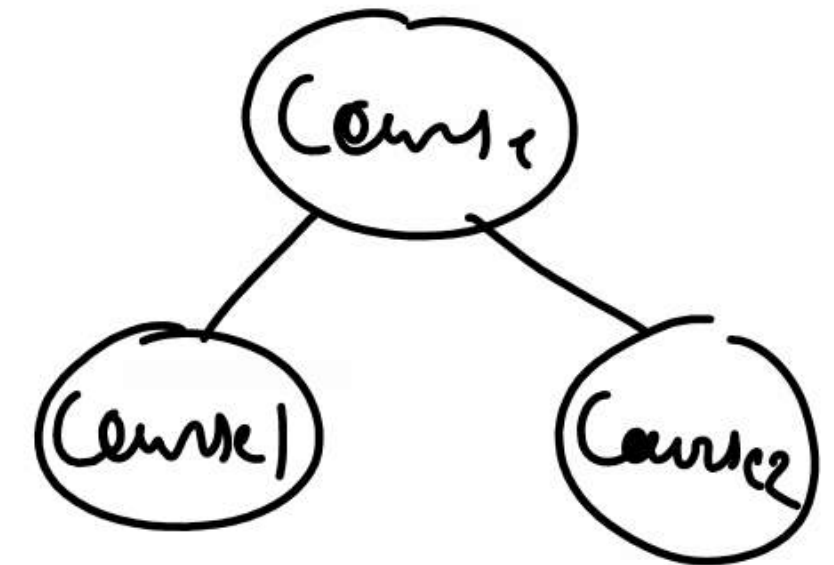


Topic :1-NF

METHOD-2ND

Here Primary key is ROLLNO but here chance of null value comes more

ROLLNO	NAME	COURSE1	COURSE2
1	RAVI	C	C++
2	HARI	JAVA ✓	NULL ✗
3	OHM	C ✓	DBMS ✓





Topic :1-NF

METHOD- 3RD (BEST WAY TO SOLVE 1 NF)

PK ←

ROLLNO	NAME
1 ✓	RAVI ✓
2 ✓	HARI ✓
3 ✓	OHM ✓

PK

BASE TABLE ----> PRIMARY KEY -ROLL NO

FK

ROLL NO	COURSE
1 ✓	C ✓
1 ✓	C++ ✓
2 ✓	JAVA ✓
3 ✓	C ✓
3 ✓	DBMS ✓

COMPOSITE KEY-(ROLLNO+COURSE)

FOREIGN KEY- ROLL NO



Topic :Closer Method

Closure of an Attribute Set-

- The set of all those attributes which can be functionally determined from an attribute set is called as a closure of that attribute set.
- Closure of attribute set $\{X\}$ is denoted as $\{X\}$

Steps to Find Closure of an Attribute Set-

- 1:-Add the attributes contained in the attribute set for which closure is being calculated to the result set
- 2:-Recursively add the attributes to the result set which can be functionally determined from the attributes already contained in the result set.



Topic :Closer Method

Q1-Consider a relation R (A , B , C , D , E , F , G) with the functional dependencies-

$$A \rightarrow BC$$

$$BC \rightarrow DE$$

$$D \rightarrow F$$

$$CF \rightarrow G$$

Now, let us find the closure of some attributes and attribute sets-

$$A^+ = \{ A \}$$

$$= \{ A , B , C \} \quad (\text{Using } A \rightarrow BC)$$

$$= \{ A , B , C , D , E \} \quad (\text{Using } BC \rightarrow DE)$$

$$= \{ A , B , C , D , E , F \} \quad (\text{Using } D \rightarrow F)$$

$$= \{ A , B , C , D , E , F , G \} \quad (\text{Using } CF \rightarrow G)$$

Thus,

$$A^+ = \{ A , B , C , D , E , F , G \}$$

Note:- If any Attribute like A here closer to all attribute including itself that A attribute is possible to be Super key or Candidate Key.



Topic :Closer Method

Super Key-

- If the closure result of an attribute set contains all the attributes of the relation, then that attribute set is called as a super key of that relation.
- Thus, we can say-
“The closure of a super key is the entire relation schema.”

Example-

In the above example,

- The closure of attribute A is the entire relation schema.
- Thus, attribute A is a super key for that relation.**



Topic :Closer Method

Candidate Key-

- If there exists no subset of an attribute set whose closure contains all the attributes of the relation, then that attribute set is called as a candidate key of that relation.

Example-

In the above example,

- No other attribute contains all the attributes of the relation, except A attribute.
- Thus, attribute A is also a candidate key for that relation.



Topic : Closer Method

Question-2 Consider the given functional dependencies-

$$AB \rightarrow CD$$

$$AF \rightarrow D$$

$$DE \rightarrow F$$

$$C \rightarrow G$$

$$F \rightarrow E$$

$$G \rightarrow A$$

Which of the following options is false?

(A) $\{CF\}^+ = \{A, C, D, E, F, G\}$

(B) $\{BG\}^+ = \{A, B, C, D, G\}$

(C) $\{AF\}^+ = \{A, C, D, E, F, G\}$

(D) $\{AB\}^+ = \{A, C, D, F, G\}$



Topic : Closer Method

Option A- $\{CF\}^+ = \{C, F\}$

$= \{C, F, G\}$ (Using $C \rightarrow G$)

$= \{C, E, F, G\}$ (Using $F \rightarrow E$)

$= \{A, C, E, E, F\}$ (Using $G \rightarrow A$)

$= \{A, C, D, E, F, G\}$ (Using $AF \rightarrow D$)

Since, our obtained result set is same as the given result set, so, it means it is correctly given.

Option B- $\{BG\}^+ = \{B, G\}$

$= \{A, B, G\}$ (Using $G \rightarrow A$)

$= \{A, B, C, D, G\}$ (Using $AB \rightarrow CD$)

Since, our obtained result set is same as the given result set, so, it means it is correctly given.

Option C- $\{AF\}^+ = \{A, F\}$

$= \{A, D, F\}$ (Using $AF \rightarrow D$)

$= \{A, D, E, F\}$ (Using $F \rightarrow E$)

Since, our obtained result set is different from the given result set, so, it means it is not correctly given



Topic :Closer Method

Option D- $\{ AB \}^+ = \{ A, B \}$

$= \{ A, B, C, D \}$ (Using $AB \rightarrow CD$)

$= \{ A, B, C, D, G \}$ (Using $C \rightarrow G$)

Since, our obtained result set is different from the given result set, so, it means it is not correctly given.

Thus,

Option (C) and Option (D) are correct.

Qusetion3:- Find the closure and candidate key

Consider a relation $R (A, B, C, D, E, F, G)$ with the functional dependencies-

- $A \rightarrow BC$
- $BC \rightarrow DE$
- $D \rightarrow F$
- $CF \rightarrow G$



Topic :Closer Method

Closure of attribute D-

$$D^+ = \{ D \}$$

$$= \{ D, F \} \quad (\text{Using } D \rightarrow F)$$

We can not determine any other attribute using attributes D and F contained in the result set.

Thus,

$$D^+ = \{ D, F \}$$

Note:- D attribute is not closer all other attribute so D attribute will not be candidate key.

Closure of attribute set {B, C}-

$$\{ B, C \}^+ = \{ B, C \}$$

$$= \{ B, C, D, E \} \quad (\text{Using } BC \rightarrow DE)$$

$$= \{ B, C, D, E, F \} \quad (\text{Using } D \rightarrow F)$$

$$= \{ B, C, D, E, F, G \} \quad (\text{Using } CF \rightarrow G)$$

Thus,

$$\{ B, C \}^+ = \{ B, C, D, E, F, G \}$$

NOTE:- Set of {B,C} Attribute is not closer to all other attribute so it will not be candidate key



Topic :Closer Method

Q4: Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M, N\}$ and the set of functional dependencies $\{\{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, K \rightarrow \{M\}, L \rightarrow \{N\}\}$ on R . What is the key for R ?

A. $\{E, F\}$

B. $\{E, F, H\}$

C. $\{E, F, H, K, L\}$

D. $\{E\}$



Topic :Closer Method

Finding attribute closure of all given options, we get:

$$\{E, F\}^+ = \{EFGIJ\}$$

$$\{E, F, H\}^+ = \{EFHGIJLMN\}$$

$$\{E, F, H, K, L\}^+ = \{EFHGIJLMN\}$$

$$\{E\}^+ = \{E\}$$

$\{EFH\}^+$ and $\{EFHKL\}^+$ results in set of all attributes, but EFH is minimal. So it will be candidate key. So correct option is (B).



Topic : Closer Method

Q5->Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys of R ?

- (a) AE, BE
- (b) AE, BE, DE
- (c) AEH, BEH, BCH
- (d) AEH, BEH, DEH



Topic : Closer Method

$(AE)^+ = \{ABECD\}$ which is not set of all attributes. So AE is not a candidate key.
Hence option A and B are wrong.

$(AEH)^+ = \{ABCDEH\}$

$(BEH)^+ = \{BEHCDA\}$

$(BCH)^+ = \{BCHDA\}$ which is not set of all attributes. So BCH is not a candidate key. Hence option C is wrong.

So correct answer is D.

THANK YOU